

Achieving Climate Neutrality: Analyzing EU Clean Energy Efforts, Innovations, and Investments

Introduction

What are the real innovation trends behind the EU's green transition, and do current clean energy investments align with the ambitious climate goals set by policymakers? Despite climate change being a prevalent discussion topic and a pressing issue ^{1,2}, a critical gap remains in understanding whether actual R&D spending and startup activity match the narrative. This study explores these dynamics by analyzing investment trends and ecosystem development in clean energy within the EU. The core hypothesis is that, although climate targets dominate political rhetoric, real innovation activity remains limited, underfunded, and geographically concentrated. To test this, we examine who is investing in clean energy R&D, where innovation is happening, and whether these hubs are positioned to lead the global transition.

Main Objectives of This Study

This study has two main goals:

- Analyze the EU's climate goals and clean energy technology efforts with a specific focus on R&D investment by major EU energy companies.
- Identify Europe's most significant clean energy innovation hubs, especially those with strong clustering around top universities, supportive institutions, and private investment.

These goals aim to assess whether innovation is truly driving the green transition, or if there is a disconnect between goals and execution.

Innovation Gaps in Clean Energy R&D

While political ambitions are high, clean energy is still not a central focus for most major corporate R&D investors. According to the European Commission's Industrial R&D Investment Scoreboard, of the 2500 largest global corporate R&D investors, only a small fraction invest significantly in clean energy.³ The vast majority of investment continues to flow into sectors like software, hardware, and automotive. For the purpose of this study only EU companies were analyzed from the top 2500 global investors.

Similarly, scaleup activity is modest. Dealroom data shows that out of almost 21000 European scaleups, fewer than 1000 belong to the clean energy category. Most of these are in early growth stages and receive relatively limited funding. This suggests that despite the climate policy spotlight, actual innovation remains peripheral and fragmented.⁴

Reaching Price/Performance Parity (P3)

Some experts, such as Robert D. Atkinson, Robin Gaster and Ed Rightor from the Information Technology & Innovation Foundation (ITIF), argue that relying solely on regulation and subsidies is unsustainable. Their report emphasizes the concept of

price/performance parity (P3), where green technologies must match or outperform fossil fuels without subsidies to achieve widespread adoption.⁵

Achieving P3 is essential for scaling technologies globally. Public funding alone is insufficient to meet projected investment needs—ranging from \$84 to \$275 trillion by 2050.⁵ The clean energy share of global GDP must rise significantly from the 1.7% recorded in 2023.⁶ Technologies that do not meet P3 will struggle to be deployed globally, especially in developing countries where affordability and reliability are paramount.

P3 includes performance metrics like lifecycle cost, usability, reliability, and scalability. For instance, electric vehicles must be as cost-effective and convenient as internal combustion vehicles. Solar must be as reliable as coal. Without P3, green technologies will rely on state support, which is not scalable globally.

Governments, therefore, must focus on supporting innovations with a clear path to P3, while markets will handle scale-up and global diffusion. Consequently, the EU should as well focus on funding and fostering innovation, instead of subsidies and regulation. These are useful tools once a technology that has the potential of becoming P3 has been identified and proven in small scale, but shouldn't be used on technologies that are not set to ever achieve P3. Enabling infrastructure—grids, storage, carbon capture—is equally essential and requires its own investment.⁵

Focus on Innovation and Investment Transparency

The EU's European Green Deal Investment Plan aims to mobilize €1 trillion over a decade. Supplementary funds like the Innovation and Modernisation Funds are expected to provide another €25 billion from ETS revenues. Yet, annual investment needs are estimated at €260 billion—raising concerns about underfunding.⁷

Institutions such as Bruegel and Cleantech for Europe have criticized the lack of transparency and unclear allocations in the EU's funding mechanisms. Bruegel notes that cost estimates may be unrealistic⁸, while Cleantech for Europe calls for a shift from regulation to targeted implementation.⁹

More concerning is the lack of support for breakthrough innovation. Despite references to innovation, few tangible outcomes have emerged. No clear EU success story exists in clean energy innovation on the scale of global leaders.¹⁰

Economic and Strategic Implications

Achieving climate neutrality is not just a cost but also an opportunity. The IMF projects that strong climate action could add 7% to global GDP by 2050. Failure to act could cost the global economy 10%, especially in vulnerable regions like South-East Asia and South America.¹¹

For the EU, this transition can reduce fossil fuel dependency, stabilize energy prices, and improve energy security—urgent needs highlighted by the war in Ukraine. Innovation-led growth could enhance regional cohesion, create skilled jobs, and boost geopolitical influence.

Clean energy innovation ecosystems will also attract ESG-aligned investors and accelerate deployment. However, this requires a shift in focus—from policy signaling to actionable innovation support and scale-up readiness.¹²

Despite the prominence of climate goals in political discourse, actual innovation and investment in clean energy remain insufficient. Most R&D funding still flows into unrelated sectors, and clean energy scaleups are few and underfunded. Only a few European regions show the right mix of ecosystem ingredients for long-term success.

To lead the global green transition, the EU must go beyond policy ambition. It must support breakthrough technologies that can reach price/performance parity and scale globally. This requires transparent, targeted investment and support for emerging innovation hubs. Without it, the EU risks falling short of its climate neutrality ambitions and missing the economic and geopolitical opportunities that innovation leadership can offer.

Methodology and data

As it was mentioned earlier, our first objective was to analyse top energy companies based on clean energy R&D investments. We used the database of the Joint Research Centre (JRC) of the European Commission, containing data on the top 2500 companies of the world based on R&D investments. The data spans 10 years, from 2014 to 2023, has the name of each company, their industry, R&D investments in euros, and several other economic factors of each. We filtered for three industries which were relevant for our study, namely alternative energy, electricity, and Gas, Water & Multiutilities and continued the analysis with companies from these sectors in each region.

The dataset used in this study for scaleups comes from Dealroom.co, a prominent global platform that offers detailed insights into startups. It provides extensive data on venture-backed startups worldwide, broken down by criteria such as location, industry, technology, funding, founders, investors, and market valuation. Since we wanted to analyse companies in a more mature stage, we only looked at scaleups, known as startups that have secured over 1 million euros in funding. Using this criterion, we extracted data from 2023 containing 20831 scaleups from Europe.

To explore the spatial distribution of these scaleups, we adopted the EU-OECD framework for Functional Urban Areas (FUAs), as outlined by OECD.¹³ FUAs include a core city and its surrounding commuting zone, offering a comprehensive view of urban economic and functional dynamics. A key benefit of using FUAs is the availability of consistent regional economic data, such as GDP, employment, and population statistics.

For filtering the data and analysing it we used Python to filter for EU scaleups and identifying scaleups which are in the field of clean energy. This was done based on the industry and the tags associated with the scaleup. The industry we could filter for was ‘energy’ and since we had to narrow this down, we used the tags associated with each scaleup to identify clean energy related ones. Starting from 916 scaleups from the ‘energy’ industry, using Python we extracted each unique tag from the scaleups and labeled each clean or not clean with the help

of natural language processing. Finally, we filtered for EU countries and remained with 500 scaleups which fit our criteria.

Table 1: Selected tags for scaleups after filtering

Lithium ion batteries	Recycling	Circular economy
Sustainable development goals	Climate tech	EV battery
EV battery recycling	Battery management system	Wind energy
Carbon capture and storage	Energy infrastructure project	EV charging
Electric mobility	EV charging infrastructure	Heat pump
Hydrogen	Hydrogen novel energy	Next-gen lithium novel energy
Solar energy	Carbon utilisation	Stationary energy storage
Other chemistry novel energy	Alternative battery chemistry	Nuclear fission
Biofuel	Smart grid	Off-grid
Building energy efficiency	Energy trading	Green buildings
Residential solar	ESCO	Geothermal energy
Marine energy	Wave energy	Biomass
Energy tech	EV routing	EV charging software
Smart home energy management	Grid energy storage	Waste heat recovery
Carbon tracking & offset	Carbon removal	EV battery production
Energy management system	Sustainable packaging	Thermal energy storage
Vibration energy harvesting	Waste to energy	Biochar
Bioplastic	Smart charging	Sustainable aviation fuel
Sustainable cement	Green buildings	Building energy retrofit
Smart meter	Direct air capture	Zero emission fuel
Large scale energy storage	Portable energy storage	Virtual power plant

For further analysis, we had to identify the investors location. This was done with the help of Python packages such as folium, and artificial intelligence tools like Copilot to retrieve coordinates based on scaleups' names. A similar method was used to identify the location of top universities and supportive institutions like accelerators and incubators. For scaleups, the coordinates were given in the Dealroom database.

Results

Clean energy R&D investments

An initial look at clean energy investments by the world's top 2,500 R&D investors reveals that EU-based companies are leading the way in this sector. As *Table 2* shows, the EU has 18 companies in the top 2500, while China, the US, and the Rest of the World have 24 collectively. The average investment from each company is the highest in the EU as well, with €256.5 million, followed by €210.1 million from China, €173 million from the US, and €144.1 million from the rest of the world. This data is from the JRC's 2023 assessment, for the 2022 year.

Table 2: Clean energy investors in the top 2500 R&D investor companies, by region (2023)

Region	Number of companies in top 2500	Average investment (€ million)	Investment range (€ million)
EU	18	256.5	1271 - 55
US	5	173.0	267.8 - 105.8
China	9	210.1	491.7 - 73.2
Rest of the World	10	144.1	561.6 – 61.3
Global	42	209.9	1271 - 55

By looking at how these values have changed over the years, we can identify some interesting trends. Shown in *Figure 1*, it's clear that over the last decade the EU companies have been always the top investors in this sector. Although changes in the absolute amount which was invested compared to other regions has changed. In 2013 the EU was already the highest investor region, and after a slight decrease in investments, since 2019 R&D investments of clean energy companies have increased even more, reaching almost €5 billion collectively.

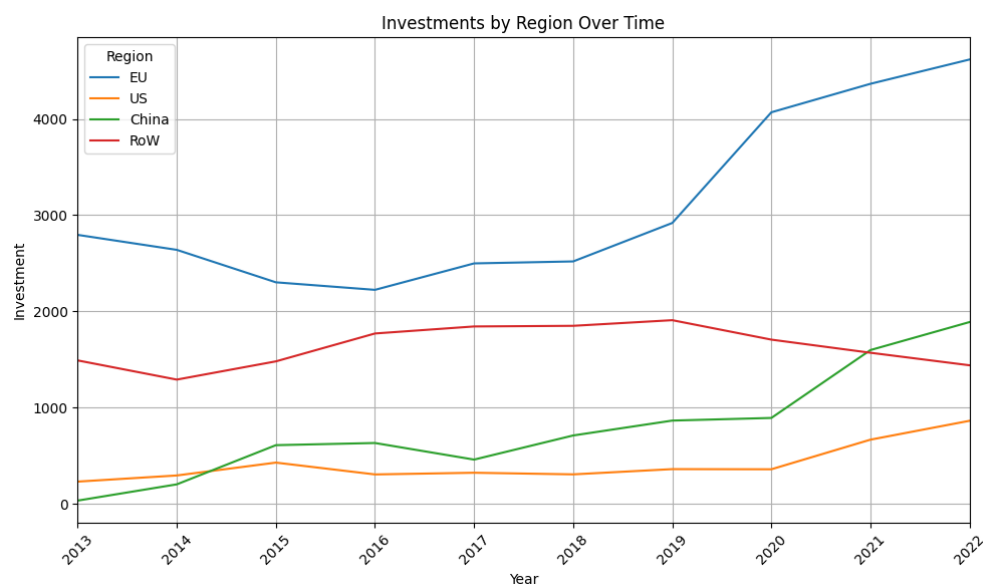


Figure 1

The significant increase in investments since 2019 could well be attributed to the introduction of the EU's Green Deal. EU based companies have started investing more and more into R&D, showing a positive impact of the Green Deal.

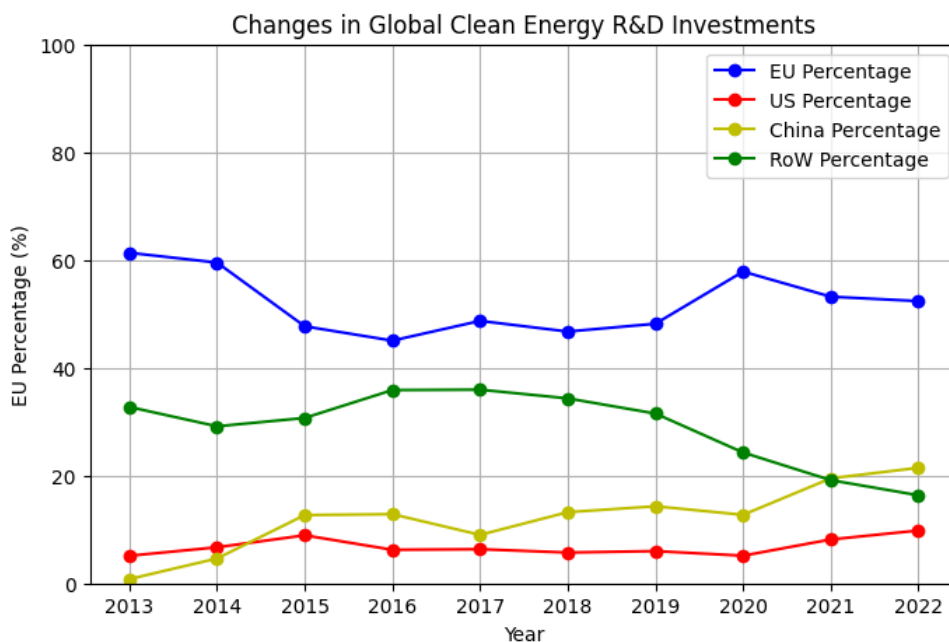


Figure 2

If we look at the share of investments by region over the years in *Figure 2*, the EU's lead is still significant, its share having been over 40% constantly, in some years reaching almost 60% as well. From both figures it can be seen that in the last few years the US and China both started to increase their investments, but China is showing faster growth in this aspect. The decline of the rest of the world can be attributed to the loss of some companies from the top 2500. While in 2019, in their top investment year they had 16 companies in the top 2500, only 10 remained in 2022.

In the EU, the top investor companies are based in France, Germany, Denmark and Spain with companies such as Electricite de France (France), Siemens Energy (Germany), Vestas Wind Systems (Denmark), and IBERDROLA (Spain).

As shown in *Figure 3*, the EU exhibits the highest average R&D investments among the regions, as shown by the position of the red diamond. It also shows the widest spread of data, with several extreme outliers—one exceeding EUR 1.2 billion. The median is noticeably lower than the mean, indicating a right-skewed distribution. This suggests that a few large companies are making disproportionately high investments, pulling the average upward.

China has a relatively high average as well, with the median and mean located close to each other. This indicates a more balanced distribution of R&D investments among companies, with fewer extreme values compared to the EU. The US follows a similar pattern to China but with a slightly lower average investment. The proximity between its median and mean also points to a stable distribution of R&D spending.

The Rest of the World (RoW) displays the lowest median and one of the lowest average investment levels. While the majority of companies in this group invest less than EUR 200 million, a few outliers—such as one close to EUR 560 million—demonstrate that there are still some high investors in this category.

Overall, the EU stands out for its high average investment and large variation, indicating that R&D efforts may be concentrated among a few dominant firms. China and the US maintain relatively steady and mid-to-high levels of R&D activity with more uniform distributions. RoW appears to lag in general, though it includes individual firms with notable investment levels. The differences between the mean and median in some regions, especially the EU, highlight the potential for policy focus on supporting broader innovation rather than relying on a few major contributors.

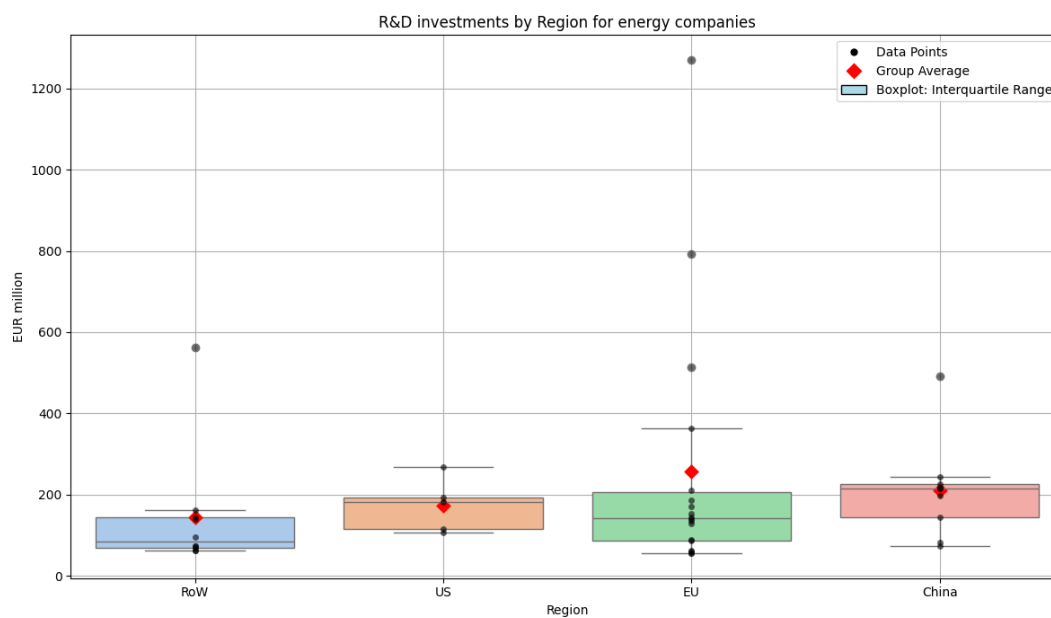


Figure 3

EU clean energy scaleup landscape

Over the past few years, the EU clean energy scaleup ecosystem has undergone significant transformation as shown in *Figures 4, 5 and 6*. Between 2021 and 2023, the number of clean energy scaleups nearly doubled, growing from just over 500 companies to almost 950. In parallel, average funding per company also saw a notable rise, from approximately €16 million in 2021 to more than €25 million in 2023. These shifts point to growing investor confidence and suggest that clean energy innovation is gaining traction within Europe's private sector.

A breakdown by company stage shows some interesting structural changes as well. Early growth scaleups now represent more than 50% of all companies, while the share of seed-stage startups has steadily decreased. At the same time, late-stage scaleups have increased their share from 22% to 27%. This evolution may reflect a natural maturation of the ecosystem, as successful early-stage companies progress and secure additional funding to scale. However, it also raises questions about the balance of support across development stages, especially in light of the EU's ambition to remain at the forefront of global clean energy innovation.

While the current trajectory is encouraging, the long-term success of the EU's clean energy transition depends not only on scaling existing solutions but also on cultivating the next generation of breakthrough technologies. An overemphasis on later-stage companies could

lead to missed opportunities if earlier-stage innovators struggle to find the resources needed to grow. Maintaining a healthy pipeline—from seed to scale—is crucial to ensure that today’s climate solutions continue to evolve and that tomorrow’s breakthroughs can emerge.

In short, the rapid growth of scaleups and rising funding levels are clear signs of momentum within the EU clean energy sector. But sustaining that momentum will require continued strategic support, particularly for early innovation, if the EU hopes to meet its climate goals and position itself as a global leader in clean technology development. For now the EU scaleup ecosystem does not reflect the emphasized importance that the EU is giving it, and it does not receive enough funding for it to help reach the EU’s ambitious climate goals. Clean energy scaleups in 2023 received, in total, more than 24 billion EUR funding, representing 9.9% of the total funding the scaleups received. In contrast, sectors like software and health received 36 and 39 billion EUR, representing 14.9% and 16.5% of the total funding. More funding would be needed in following years for clean energy projects to achieve growth and for new players to be able to enter the scene.

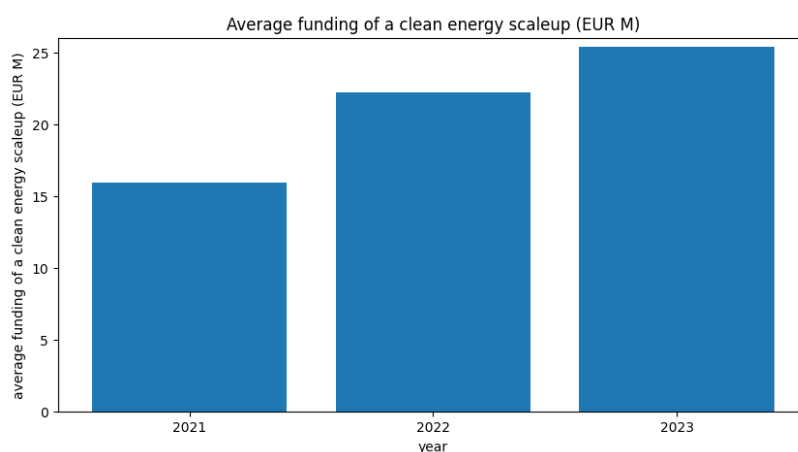


Figure 4

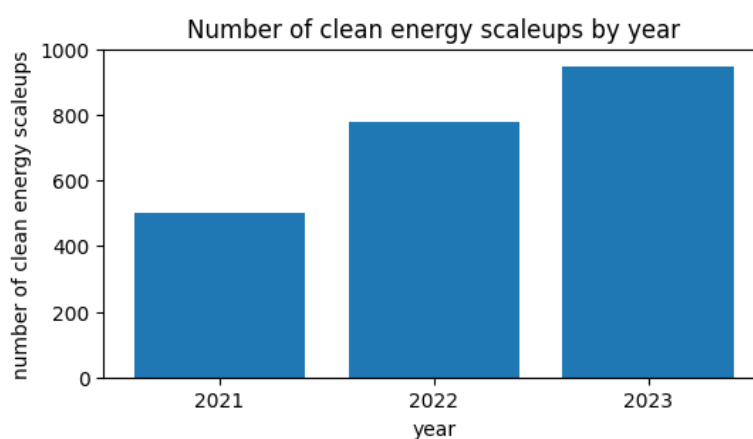


Figure 5

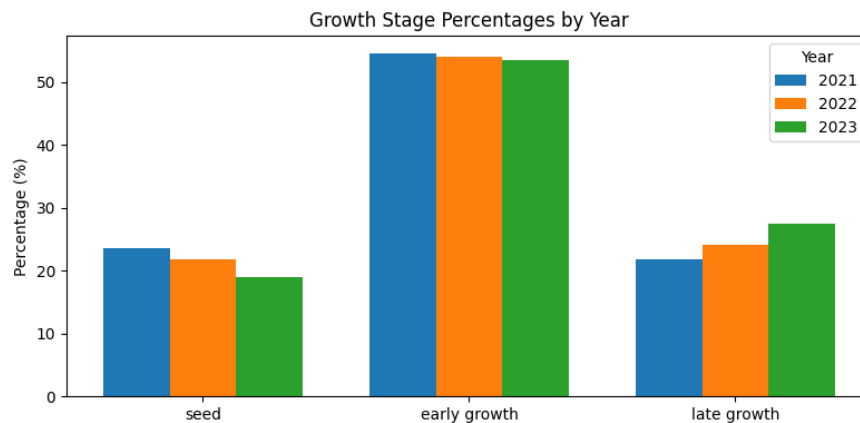


Figure 6

Scaleups do not appear randomly. They tend to emerge in environments where key enabling conditions are present—namely, access to talent, capital, infrastructure, and institutional support. According to Engel, one of the core characteristics of successful entrepreneurial ecosystems is the close proximity of top-tier research universities, supportive institutions such as accelerators and incubators, and a well-developed risk capital market.¹⁴ These elements create a feedback loop that helps innovative companies not only launch, but also scale and thrive over time.

In the clean energy sector, this dynamic is particularly important. Clean energy scaleups often require more than just strong business ideas—they need access to cutting-edge research, specialized facilities, regulatory guidance, and long-term investment to navigate the capital-intensive path of developing and deploying new technologies. Unlike software startups, which can often scale rapidly with minimal infrastructure, clean energy companies frequently face significant barriers to entry, from prototype development to regulatory compliance and grid integration. This makes their reliance on ecosystem support even more pronounced.

To identify the most important scaleup hubs in the EU, we analyzed the geographic distribution of clean energy scaleups alongside the locations of four key ecosystem components: (1) leading research universities, which serve as sources of both innovation and talent; (2) investors, particularly those focused on early-stage and clean tech ventures; (3) incubators and accelerators, which help companies navigate early growth stages; and (4) policy and institutional frameworks that create favorable conditions for green innovation. This combined analysis helps highlight regions where these elements converge and where the ecosystem is most conducive to producing globally competitive clean energy scaleups.

By focusing on these indicators, it becomes easier to distinguish between regions where scaleups appear by chance, and those where they are systematically supported. Understanding this distinction is essential not only for mapping the current innovation landscape, but also for identifying where future investments, policy efforts, or institutional support might be most effectively directed. Knowing this, we mapped the 947 scaleups identified earlier using their coordinates on *Figure 7*.

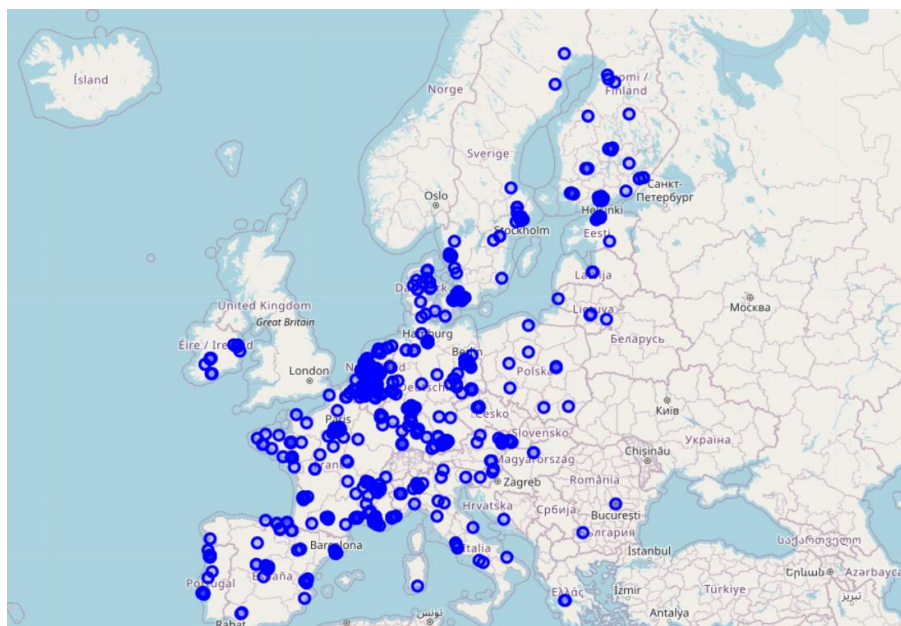


Figure 7: The 947 EU clean energy scaleup from 2023

In order to better analyse and identify the most important hubs, we clustered everything to FUAs, as it was mentioned earlier. We did this with the scaleups as well on *Figure 8*.

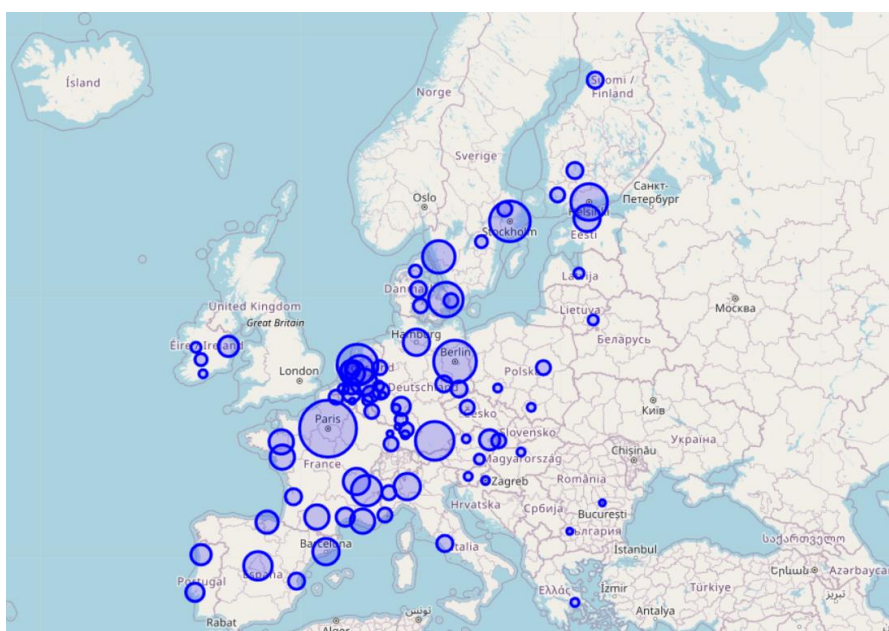


Figure 8: EU clean energy scaleups clustered to FUAs

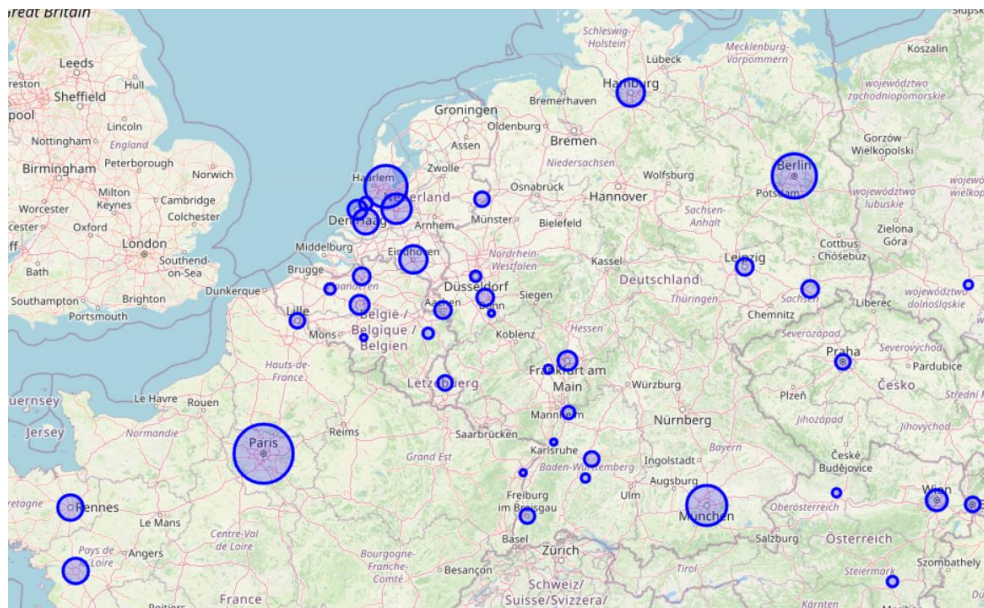


Figure 9: Clean energy scaleups clustered to FUAs, zoomed in

As it can be seen on *Figures 8 and 9*, the FUAs where the most scaleups concentrate are mostly in Western and Northern Europe. As we move more towards Eastern Europe, clean energy scaleups become less and less frequent. This regional disparity may be attributed to a combination of factors, including lower levels of private and public funding for innovation, weaker institutional support, and less mature entrepreneurial ecosystems. Additionally, regulatory uncertainties, limited access to international networks, and brain drain may also hinder the growth and scaling of clean energy startups in these regions. The imbalance highlights the need for targeted policy interventions and investment incentives to stimulate innovation and bridge the gap between Western/Northern and Eastern Europe in the clean energy transition.

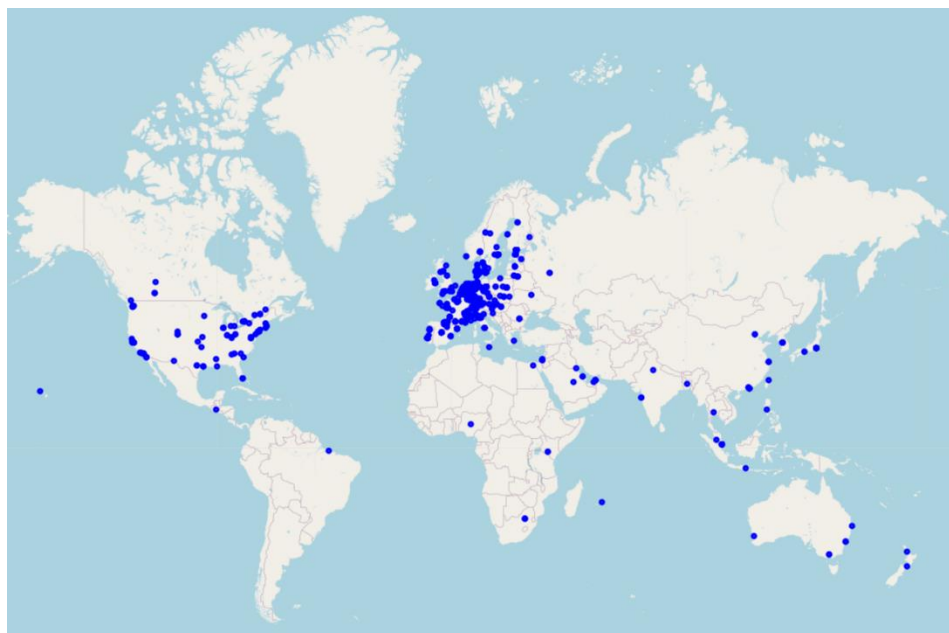


Figure 10: Clean energy scaleup investors



Figure 11: Clean energy scaleup investors from the EU, clustered to FUAs

On Figures 10 and 11, we mapped the investors of these 947 scaleups, first globally, then focusing only on the EU. Over 72% of the investors were from the EU, the major investor from outside the EU being the US. If we look at the EU investors, most are from western and northern FUAs as well, like Paris, Berlin, Stockholm, Helsinki or Amsterdam.



Figure 12: EU accelerators and incubators, clustered to FUAs (14)

Taking a look at the top 100 supportive institutions from the EU, a more evenly distributed picture appears. The most influential FUAs in this context are still in France, Germany and Spain, so mostly in Western Europe, but there are significant hubs in Dublin, Baltic capitals, Portugal and with the most from Eastern Europe, Bucharest.



Figure 13: Top 100 EU university, clustered to FUAs (15)

The top 100 universities from the EU are, again, almost entirely concentrated in Western and Northern Europe, reflecting investments in education, research infrastructure, and international collaboration. Countries like Germany, the Netherlands, Sweden, and France consistently dominate these rankings, while institutions in Southern and Eastern Europe are less represented, often due to differences in funding, academic resources, and global visibility.

Based on these aspects we could determine the most important EU clean energy scaleup hubs, considering all criteria presented. Ranking was done, for each FUA, by normalizing the number of these in each category between 0 and 1, then the four categories were added together (scaleups, supportive institutions, universities and investors). The most important clean energy scaleup hubs of the EU determined this way were Paris, Berlin, Madrid, Helsinki and Amsterdam. They had the best numbers in all categories, meaning that these are the cities in the EU where investors, supportive institutions and universities are all present and because of this these are the cities where also the most clean energy scaleups are located.

Concluding remarks

This study has shown that while the EU is at the forefront of clean energy R&D investment and is home to a growing number of clean energy scaleups, there are still several structural weaknesses that must be addressed if the EU is to become a true global leader in the green transition. The current concentration of innovation and investment in Western and Northern Europe underscores a persistent regional imbalance that risks leaving much of Eastern and Southern Europe behind. Similarly, although overall investment figures are promising, the

lack of clarity around long-term strategy, measurable outcomes, and targeted innovation support limits the effectiveness of the EU's green policies.

If the EU is to maintain momentum and reach its climate neutrality targets, a stronger focus must be placed on enabling breakthrough technologies through more detailed and ambitious support frameworks. The transition will not be achieved by subsidies and regulation alone. Without market-competitive clean technologies, capable of outperforming fossil fuels on their own merits, long-term climate goals will remain out of reach. The EU must therefore prioritize innovation ecosystems, address regional disparities, and ensure that early-stage clean energy innovators are not left behind in favor of already mature scaleups. The success of the global energy transition hinges on solutions that are not only effective and scalable, but also equitable and inclusive across all the regions of the world.

Bibliography

1. IPCC. (2023). *Summary for Policymakers*. In *Climate Change 2023: Synthesis Report. Contribution of Working Groups I, II and III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* (Core Writing Team: H. Lee & J. Romero, Eds.). Geneva, Switzerland: IPCC. <https://doi.org/10.59327/IPCC/AR6-9789291691647.001>
2. Energy Institute. (2024). *Statistical Review of World Energy 2024*. Retrieved from <https://www.energyinst.org/statistical-review>
3. Nindl, E., Confraria, H., Rentocchini, F., Napolitano, L., Georgakaki, A. et al., 2023 *EU industrial R&D investment scoreboard*, Publications Office of the European Union, 2023, <https://data.europa.eu/doi/10.2760/506189>
4. Dealroom data
5. Atkinson, R. D., & McCormick, L. (2023, July 10). *Beyond Force: A Realist Pathway Through the Green Transition*. Information Technology & Innovation Foundation. <https://itif.org/publications/2023/07/10/beyond-force-a-realist-pathway-through-the-green-transition/>
6. BloombergNEF. (2024). *Energy Transition Investment Trends 2024*. Retrieved from <https://about.bnef.com/clean-energy-investment/>
7. European Commission. (2020). *European Green Deal Investment Plan*. Retrieved from https://ec.europa.eu/commission/presscorner/detail/en/fs_20_48
8. Pisani-Ferry, J., & Tagliapietra, S. (2024). *An investment strategy to keep the European Green Deal on track* (Policy Brief No. 31/2024). Bruegel. <https://www.bruegel.org/policy-brief/investment-strategy-keep-european-green-deal-track>
9. Cleantech for Europe. (2024). *Policy update: The EU Green Deal's legacy*. Retrieved from <https://www.cleantechforeurope.com/policy/policy-update-the-eu-green-deals-legacy>

10. Cleantech for Europe. (2024). *Cleantech Reality Check: The State of EU Cleantech Innovation*. Retrieved from <https://www.cleantechforeurope.com/insight/cleantech-reality-check>
11. International Monetary Fund (IMF). (2023, December 5). *Benefits of Accelerating the Climate Transition Outweigh the Costs*. Retrieved from <https://www.imf.org/en/Blogs/Articles/2023/12/05/benefits-of-accelerating-the-climate-transition-outweighthe-costs>
12. Swiss Re Institute. (2021). *The Economics of Climate Change: No Action Not an Option*. Retrieved from <https://www.swissre.com/institute/research/topics-and-risk-dialogues/climate-and-natural-catastrophe-risk/expertise-publication-economics-of-climate-change.html>
13. Dijkstra, L., Poelman, H., & Veneri, P. (2019). *The EU-OECD Definition of a Functional Urban Area* (OECD Regional Development Working Papers No. 2019/11). OECD Publishing. <https://doi.org/10.1787/d58cb34d-en>
14. Engel, J. S. (Ed.). (2014). *Global Clusters of Innovation: Entrepreneurial Engines of Economic Growth around the World*. Edward Elgar Publishing.